

001 100. The speed of the motor can be changed by applying variable delays between the pulses applied to it. This is done by the `speedf()` function. As per the user's choice, this function inserts a delay in between the pulses. As a result, the pulse repetition frequency varies and the motor runs at different speeds.

Construction

You can assemble this project on any general-purpose PCB or on the PCB

shown here in Fig. 5. The component layout for the PCB is shown in Fig. 6. After mounting all the components on the PCB, fix the 25-pin male connector (CON1) on the PCB as shown in the layout. Use a 4-wire, 50cm FRC cable with 25-pin female connector on one end and male connector on the other end for connection between the board and the PC. Before connecting the stepper motor in the circuit, first identify the coils by using a multimeter set in 'ohms' range. The

resistance between the common wire and an individual coil is about 2.5 ohms and the coil-to-coil resistance is about 5 ohms. Connect the common wire to 5V and the rest of the coils in sequence as shown in the schematic diagram.

EFY note. The source code and other relevant files of this article have been included in this month's EFY-CD.

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SMC_V1.0.C

```

/* Source filename : SMC_V1.0.c */
#include<stdio.h>
#include<conio.h>
#include<graphics.h>
#include<dos.h>
void cw(); //Function declaration for clockwise direction
void acw(); //Function declaration for anticlockwise direction
void speedf(); //Function declaration for speed control
unsigned int i,j,m,s; //All variables declaration
char c;
void main() //Main program begins here
{
//Initialize the graphics mode
int graphdriver = DETECT, graphmode;
initgraph(&graphdriver, &graphmode, "c:\\tc\\bgi");
cleardevice();
settextstyle(1,HORIZ_DIR,2);
setbkcolor(1);
setcolor(LIGHTRED);
rectangle(70,35,580,405);
rectangle(75,40,575,400);
outtextxy(115,70,"welcome to stepper motor controller v1.0");
outportb(0x378,0); //Clear parallel port by sending all 0s
sleep(3);
do
{
cleardevice();
setbkcolor(1);
setcolor(GREEN);
rectangle(70,35,580,405);
rectangle(75,40,575,400);
rectangle(143,72,506,92);
rectangle(145,74,504,90);
settextstyle(1,HORIZ_DIR,1);
outtextxy(148,70,">>>STEPPER MOTOR CONTROLLER V1.0<<<");
settextstyle(0,HORIZ_DIR,1);
outtextxy(150,100,">>>1. ENTER THE NO. OF FULL-ROTATIONS TO RUN=");
gotoxy(63,7);
scanf("%d",&m);
outtextxy(150,116,">>>3. ENTER 'a' FOR ACW AND 'c' FOR CW=");
gotoxy(58,8);
scanf("%s",&c);
outtextxy(150,132,">>>4. ENTER THE RE-
QUIRED SPEED b/w(1-3)=");
gotoxy(58,9);
scanf("%d",&s);
setcolor(LIGHTRED);
outtextxy(190,165,">>> Press Any Key to Run.....");
getch();
if((c=='a')||(c=='A'))
{
acw();
outportb(0x378,0);
setcolor(LIGHTRED);
outtextxy(150,220,"STOPPED.<<<Press any key to Restart Or Esc to EXIT...");
}
else
if((c=='c')||(c=='C'))
{
cw();
outportb(0x378,0);
outtextxy(150,220,"STOPPED.>>>Press any key to Restart Or Esc to EXIT...");
}
}while(getch() != 27);
setcolor(LIGHTRED);
outtextxy(150,240,">>>PROGRAM BY:-");
outtextxy(150,260,">>>Harnam-Singh, b-tech, e.c.e, ptk (pb)");
outtextxy(150,280,">>>Email-> e_r_harnam@yahoo.com");
outtextxy(150,300,">>>THANK YOU FOR USING THIS PROGRAM...!");
sleep(3);
}
void cw()
{
sound(1300);
outtextxy(150,196,"<<<RUNNING-CLOCKWISE-");
for(i=1;i<=m;i++)
{
for(j=1;j<=16;j++)
{
outportb(0x378,1);
speedf();
gotoxy(41,13);
printf("001->");
outportb(0x378,2);
speedf();
gotoxy(41,13);
printf("010->");
outportb(0x378,4);
speedf();
gotoxy(41,13);
}
}
}
void acw()
{
sound(1300);
outtextxy(150,196,"<<<RUNNING-ANTICLOCK-");
for(i=1;i<=m;i++)
{
for(j=1;j<=16;j++)
{
outportb(0x378,2);
speedf();
gotoxy(45,13);
printf("<-010");
outportb(0x378,1);
speedf();
gotoxy(45,13);
printf("<-100");
outportb(0x378,4);
speedf();
gotoxy(45,13);
printf("<-100");
}
}
}
void speedf()
{
if(s==1)
{
delay(400);
}
else if(s==2)
{
delay(300);
}
else if(s==3)
{
delay(200);
}
else if(s>3)
{
delay(100);
}
}
printf("100->");
}
gotoxy(47,13);
printf("%d Rotation(s)",i);
}
nosound();
}
}
}

```